Business Requirement Document (BRD)

Project Name: BNPL Risk Scoring & Analytics (India)

By: Tejas Chopade (Data Science)

---

1. Introduction

The purpose of this document is to define the business and technical requirements for developing a **Buy Now Pay Later (BNPL) Risk Scoring & Analytics** focuses on the Indian fintech ecosystem.

The system aims to:

- Analyze customer behavior.
- Predict default risk.
- Enable smarter credit decision-making.

This project aligns with modern fintech practices used by companies like LazyPay and Simpl.

---

2. Business Objectives

- **Reduce loan default rates**: Identify high-risk users early using predictive models. Which helps in minimizing financial losses and improving lending efficiency.
- **Improve credit risk assessment**: Use data-driven scoring instead of rule-based decisions. Enable more accurate and scalable credit evaluation.
- **Enable customer segmentation**: Group users based on behavior, income, and repayment patterns. Helps in target offers and manage risk across segments.
- **Optimize BNPL credit limits**: Analyze risk profile based on credit limits. Which prevents over-lending while maximizing customer usage.
- **Provide data-driven insights for growth**: Use analytics to understand trends and user behavior. Which supports strategic decisions for business expansion.

---

3. Scope

- Data ingestion from structured datasets.
- Data cleaning & preprocessing.
- Feature engineering.
- EDA & statistical analysis.
- SQL-based analytics.
- Dashboard & reporting.
- Deployment.

---

4. Stakeholders

- Data Scientists.
- Data Analysts.
- Product Managers.
- Business Analysts.
- Fintech Companies.

---

5. Data Sources

- Synthetic BNPL dataset.
- Transaction-level datasets.
- SQL databases.
- Optional APIs.

---

6. Data Science Pipeline

- **Data Ingestion**: Collect data from multiple sources like CSV, APIs, and databases.
  Ensure availability of raw data for analysis and modeling.
- **Data Cleaning**: Handle missing values, duplicates, and incorrect entries.
  Improve data quality and reliability of results.

- **Data Transformation**: Convert raw data into structured and usable formats.
  Include aggregation, joins, and feature creation.
- **EDA (Exploratory Data Analysis)**: Analyze data using statistics and visualizations.
  Help discovering patterns, trends, and anomalies.
- **Modeling**: Build machine learning models to predict risk. Convert data insights into actionable predictions.
- **Visualization**: Present insights using dashboards and charts. Makes complex data easy to understand for stakeholders.
- **Deployment**: Deploy models and automate workflows. Enable real-time or batch predictions in production.

---

7. Data Cleaning & Preprocessing

- **Missing value imputation**: Replace missing values using mean, median, or advanced methods. Prevent data loss and improve model performance.

- **Outlier handling**: Detect extreme values using IQR or Z-score methods. Reduces noise and improves model stability.

- **Encoding categorical variables**: Convert text data into numerical format (Label/One-hot encoding). Allows machine learning models to process categorical data.

- **Normalization & scaling**: Adjust feature values to a standard range. Improves model convergence and accuracy.

---

8. Feature Engineering

Key Features:

- **Income-to-loan ratio**: Measures user's repayment capacity relative to income. Help identify financially stressed users.

- **Credit utilization ratio**: Indicates how much credit limit is being used. High utilization often signals higher risk.

- **Repayment consistency**: Tracks how regularly a user pays installments. Strong indicator of credit worthiness.

- **Transaction frequency** Measures how often a user uses BNPL services. Help identify active vs risky over-users.

- **Behavioral score**: Combines multiple user actions into a single score. Used for segmentation and risk classification.

9. Exploratory Data Analysis (EDA)

- **Descriptive statistics:** Summarize data using mean, median, standard deviation. Provide an overview of dataset distribution.
- **Distribution analysis (Gaussian):** Analyze how data is spread across values. Help detect skewness and anomalies.
- **Correlation heatmaps:** Show relationships between variables. Help identify important features for modeling.
- **Cohort analysis:** Group users based on time or behavior. Track user retention and repayment trends.

---

10. ER Diagram (Database Design)

Users (Fact Table)

- User_id (Primary Key) – Unique identifier for each user.
- Age – Age of the user.
- Monthly_income – Monthly income of the user.
- City – City of residence.
- State – State of residence.
- Reason – Purpose of loan usage.
- Default_risk – Risk score (range 1–9).

Accounts

- S_no (Primary Key) – Unique account record ID.
- User_id (Foreign Key → Users) – Links account to user.
- Lending_company – Name of BNPL provider.
- Credit_limit – Approved credit limit for the user.

Transactions

- Credit_utilisation (Primary Key) – Unique utilization metric.
- User_id (Foreign Key → Users) – Links loan to user.
- Loan_amount – Amount borrowed.
- Tenure_months – Loan duration.
- Year – Year of loan issuance.

Accounts

- Payment_id (Primary Key) – Unique ID (format: ABC12345).
- User_id (Foreign Key → Users) – Links payment to user.
- Repayment_method – Mode of repayment (UPI, Card, etc.).

- Missed_payments – Number of missed payments.
- Transactions_per_month – User activity level.

Relationships

- One **User** → Many **Accounts**.
- One **User** → Many **Loans**.
- One **User** → Many **Payments.**

---

11. SQL Analysis

- **High-risk users' identification:** Query users with high defaults or missed payments. Help prioritize risk mitigation strategies.
- **Company-wise default rate:** Compare default rates across BNPL providers. Identify which platforms have higher risk.
- **Payment method trends:** Analyze which repayment methods lead to defaults. Help improve payment collection strategies.
- **City/State analysis:** Identify regional differences in user behavior. Support location-based decision making.

---

12. Statistical Analysis

Methods:

- **Hypothesis testing (T-test, Z-test)**: Validate assumptions using statistical methods. Ensures insights are statistically significant.

- **Regression analysis**: Understand relationship between variables. Helps predict outcomes like default probability.

---

13. Visualization & Reporting

- **Risk segmentation dashboard:** Display users grouped by risk levels. Help quick identification of risky segments.
- **Company comparison dashboard:** Compare BNPL providers on key metrics. Supports competitor analysis.
- **Trend analysis dashboard:** Show growth and usage patterns over time. Help understand market evolution.

- **User behavior insights:** Visualize spending and repayment habits. Supports personalized strategies.

---

14. Competitor Analysis

Key Players: LazyPay, Simpl, Slice, ZestMoney, KreditBee & Others.

Analysis:

- **Risk strategies**: Methods used by BNPL companies to assess and control user credit risk. Includes credit scoring, behavioral analysis, and limit restrictions.
- **Target users**: Specific customer segments that BNPL companies focus on for growth. Typically includes young users, online shoppers, and thin-credit individuals.
- **Revenue models:** Ways BNPL companies generate income from their services. Includes merchant fees, late payment charges, and interest on EMIs.

---

15. KPIs & Success Metrics

- **Default rate reduction**: Measure decreases in loan defaults. Indicates effectiveness of risk model.
- **Model accuracy (>80%)**: Evaluate prediction correctness. Ensure reliability of model outputs.
- **Customer segmentation quality**: Measure clarity and usefulness of segments. Help improve targeting strategies.
- **Dashboard usability**: Evaluate ease of understanding insights. Ensures adoption by business users.

---

16. Risks & Challenges

- **Data quality issues**: Poor data can lead to incorrect insights. Requires strong preprocessing.
- **Imbalanced dataset**: Fewer default cases can bias models. Needs balancing techniques like SMOTE.
- **Model bias**: Model may unfairly favor certain groups. Requires fairness checks.

- **Lack of real BNPL data**: Synthetic data may not fully reflect reality. Need of careful simulation.

---

17. Future Enhancements

- **Real-time scoring**: Enable instant credit decisions. Improve user experience.
- **Banking API integration**: Use real financial data for better accuracy. Enhance model reliability.
- **Deep learning models**: Use advanced models for better predictions. Handle complex patterns.
- **Personalized recommendations**: Suggest offers based on user behavior. Increase engagement and retention.

---

18. Conclusions

This project delivers a **complete end-to-end BNPL analytics** that integrates multiple components of the data science lifecycle, including:

- **Data engineering** to collect, clean, and structure raw financial and behavioral data
- **Statistical analysis** to identify patterns, trends, and validate key assumptions
- **Machine learning models** to predict default risk and generate user risk scores
- **Business insights** to support strategic decision-making and customer targeting

By combining these elements, the system provides a holistic view of user behavior and credit risk, enabling fintech companies to make more accurate lending decisions, optimize credit limits, and proactively reduce financial risk and defaults.
